

Lecture 6

Symmetry, projectors, and degeneracy

Symmetry tells us the degeneracies of a Hamiltonian before we diagonalise it. This lecture shows how. We first say precisely when a group is a symmetry of a quantum system: its operators commute with the Hamiltonian. Characters then give projectors that split any representation into its isotypic components. An operator that commutes with the group is block-diagonal in a basis adapted to the symmetry, and each of its eigenvalues comes in a multiplet whose size is the dimension of an irreducible representation. We prove this once. Here we apply it to a quantum particle hopping on a triangle, and in Lecture 7 to the vibrations of a molecule.

By the end of this lecture you will be able to:

- ▶ Say when a group is a symmetry of a quantum system.
- ▶ Project a representation onto its isotypic components.
- ▶ Predict degeneracies from the block-diagonalisation theorem.

6.1 Symmetry in quantum mechanics

The state space of a quantum system is a complex vector space, and a symmetry acts on it linearly, by the function-space construction of Lecture 3:

$$(\rho(g)\psi)(\vec{x}) = \psi(g^{-1}\vec{x}).$$

For g orthogonal this preserves the norm of square-integrable functions, so ρ is a unitary representation. (Wigner's theorem, met in Part II, says that every symmetry of quantum mechanics is implemented by a unitary or an antiunitary operator; for point groups the unitary action above is all we need.)

Definition 6.1 (Quantum symmetry). Let ρ be a unitary representation of a group on the state space. The group G is a *symmetry* of the quantum system with Hamiltonian H if

$$[\rho(g), H] = 0 \quad \text{for all } g \in G.$$

In the language of Lecture 3, H is an intertwiner of the state space with itself.

For an unbounded H the commutator is meant on a common domain of smooth, *compactly supported* functions, that is, functions that vanish outside some ball of finite radius. Every $\rho(g)$ preserves this domain, because a rotation or a reflection maps a ball to a ball of the same radius. That the commutator then vanishes on the whole domain of H , eigenfunctions included, we take for granted.

Proposition 6.2. Let $H = -\frac{\hbar^2}{2m}\nabla^2 + V(\vec{x})$, with the group acting by the function-space action above. Then $G \subset O(3)$ is a symmetry if and only if the potential is invariant: $V(g\vec{x}) = V(\vec{x})$ for all $g \in G$.

The kinetic term commutes with every rotation and reflection, because the Laplacian does; so only the potential can break the symmetry. The self-study section at the end of the lecture checks the Laplacian.

Proposition 6.3. If $[\rho(g), H] = 0$ for all g , then every energy eigenspace $\{\psi : H\psi = E\psi\}$ is an invariant subspace. So each eigenspace carries a representation of G .

Proof. $H\rho(g)\psi = \rho(g)H\psi = E\rho(g)\psi$. ■

So the energy levels of a symmetric system are representations of its symmetry group. Which representations, and how large? Answering that is the business of this lecture.

In classical mechanics a symmetry maps solutions of the equations of motion to solutions, and it does no more. A finite group in particular conserves nothing: conservation laws come from *continuous* families of symmetries, by Noether's theorem, which Part II proves. Quantum mechanics gets more out of a discrete symmetry. The reason is the superposition principle: the symmetry acts on a vector space, and this lets us apply the theory of linear representations.

6.2 Projection operators

Characters compute not only multiplicities: together with the operators $\rho(g)$, they also find subspaces. From here on G is finite and V is a finite-dimensional complex representation, and we keep two notations apart: superscripts in parentheses, $V^{(i)}$, denote the abstract irreducible representations, and subscripts, V_i , the pieces of a given representation V defined next.

Definition 6.4 (Isotypic components). Let (ρ, V) be a representation. The *isotypic component* V_i of type i is the sum of all irreducible invariant subspaces of V equivalent to $V^{(i)}$.

If $V \cong \bigoplus_i m_i V^{(i)}$, the component V_i collects the m_i copies of $V^{(i)}$, so $\dim V_i = m_i n_i$. When $m_i > 1$, the individual copies inside V_i are *not* unique, but the component is: the next theorem produces it from the character and the operators $\rho(g)$, without choosing a decomposition.

Theorem 6.5 (Character projectors). For each irreducible representation define

$$P_i = \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g).$$

Then P_i commutes with every $\rho(h)$, and it acts as the identity on every irreducible invariant subspace of type i and as zero on those of every other type. So P_i is the

projection onto V_i , the P_i sum to the identity, and

$$V = \bigoplus_i V_i.$$

Proof. First, conjugate P_i by $\rho(h)$ and rename $g' = hgh^{-1}$:

$$\rho(h)P_i\rho(h)^{-1} = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \rho(hgh^{-1}) = \frac{n_i}{|G|} \sum_{g'} \overline{\chi_i(h^{-1}g'h)} \rho(g') = P_i,$$

because χ_i is a class function. Next, let W be an irreducible invariant subspace of type j . The operator P_i maps W into itself and commutes with the action on it, so by Schur's lemma it is λid_W . The trace fixes λ :

$$\lambda n_j = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \chi_j(g) = n_i \langle \chi_i, \chi_j \rangle_G = n_i \delta_{ij}.$$

Finally, write V as a direct sum of irreducible invariant subspaces by Maschke's theorem. On each summand of type i , P_i is the identity, and on every other summand it is zero. So P_i projects onto the sum of the type- i summands, and $\sum_i P_i$ is the identity on every summand. Every irreducible invariant subspace W of type i lies in the image of P_i , because P_i is the identity on W . So the image is all of V_i , and it does not depend on the decomposition chosen. ■

Example 6.6 (The three projectors of S_3 on $\mathbb{C}^3$). Take the permutation representation of $S_3 \cong C_{3v}$ on $\mathbb{C}^3$, and let J be the 3×3 matrix all of whose entries are 1. Let C be the matrix of the 3-cycle $(1\ 2\ 3)$. The three rotation matrices $\mathbb{1}$, C , C^2 have their 1s in different places and together fill J :

$$\mathbb{1} + C + C^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = J.$$

The three transposition matrices fill J in the same way. With the characters of Lecture 5,

$$P_{A_1} = \frac{1}{6}(J + J) = \frac{1}{3}J, \quad P_{A_2} = \frac{1}{6}(J - J) = 0, \quad P_E = \frac{2}{6}(2\mathbb{1} - C - C^2) = \mathbb{1} - \frac{1}{3}J.$$

The first projects onto the line $L = \mathbb{C}(1, 1, 1)$ — it is the averaging operator of Lecture 5 — and the third onto the plane V_0 of vectors whose components sum to zero. The projector P_{A_2} vanishes, which records that A_2 does not occur. The traces 1, 0, 2 are the dimensions $m_i n_i$. Applied to the first standard basis vector e_1 , the projectors split it into its two components:

$$e_1 = P_{A_1} e_1 + P_E e_1 = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}.$$

The second vector's components sum to zero, so it lies in V_0 . ◀

6.3 The block-diagonalisation theorem

Now let an operator A on V commute with the group, $A\rho(g) = \rho(g)A$ for all g . Physically, A is a Hamiltonian, or the dynamical matrix of Lecture 7, that the symmetry leaves unchanged.

Such an A commutes with every P_i , since P_i is a combination of the $\rho(g)$. So A maps each isotypic component into itself, and it has no matrix elements between different components. What does it do inside one?

Choose a *symmetry-adapted basis* of V_i : split V_i into its m_i copies of $V^{(i)}$, and in the a th copy choose vectors $w_1^{(a)}, \dots, w_{n_i}^{(a)}$ that transform by the same matrices in every copy,

$$\rho(g) w_\nu^{(a)} = \sum_\mu D_{\mu\nu}^{(i)}(g) w_\mu^{(a)}, \quad a = 1, \dots, m_i.$$

Such a basis exists: choose a basis of the first copy, and carry it to the other copies by isomorphisms of representations. With an invariant inner product (Lecture 4) it can be chosen orthonormal: take the copies mutually orthogonal and the isomorphisms unitary. Schur's lemma allows the latter, because an isomorphism T between unitary irreducibles has $T^\dagger T = c \text{ id}$ with $c > 0$, so $T/\sqrt{c}$ is unitary. The index μ is called the *partner index*, and a counts the copies.

In practice the copies are usually built alike. Suppose two sets of basis vectors are permuted by the group in the same way. The map that sends each vector of the first set to its counterpart in the second is then an isomorphism of representations, so the same linear combination, taken in each set, gives vectors that transform by the same matrices. Section 6.4 and Lecture 7 build their adapted bases this way.

Theorem 6.7 (Block diagonalisation). Let A commute with every $\rho(g)$. In a symmetry-adapted basis of V_i ,

$$A w_\nu^{(b)} = \sum_{a=1}^{m_i} F_{ab}^{(i)} w_\nu^{(a)},$$

with an $m_i \times m_i$ matrix $F^{(i)}$, the *reduced matrix*, that does not depend on the partner index ν . Consequently, if $F^{(i)}$ has an eigenvalue λ with an r -dimensional eigenspace, then the eigenspace of A for λ inside V_i is $r n_i$ -dimensional. If A is Hermitian and the basis orthonormal, $F^{(i)}$ is Hermitian.

Proof. Since A preserves V_i , write $A w_\nu^{(b)} = \sum_{a,\mu} K_{\mu\nu}^{ab} w_\mu^{(a)}$, with an $n_i \times n_i$ matrix K^{ab} for each pair a, b . The transformation law gives

$$A \rho(g) w_\nu^{(b)} = \sum_{a,\mu} (K^{ab} D^{(i)}(g))_{\mu\nu} w_\mu^{(a)}, \quad \rho(g) A w_\nu^{(b)} = \sum_{a,\mu} (D^{(i)}(g) K^{ab})_{\mu\nu} w_\mu^{(a)}.$$

The two are equal, so for each pair a, b

$$K^{ab} D^{(i)}(g) = D^{(i)}(g) K^{ab} \quad \text{for all } g.$$

So the $n_i \times n_i$ matrix K^{ab} commutes with an irreducible representation, and by Schur's lemma it is a multiple of the identity, $K^{ab} = F_{ab}^{(i)} \mathbb{1}$. This is the formula. It says that A acts on the m_i coefficients of each partner ν separately, and by the same matrix $F^{(i)}$ for every ν . So if c is an eigenvector of $F^{(i)}$ with eigenvalue λ , the n_i vectors $\sum_b c_b w_\nu^{(b)}$, one for each ν , are independent eigenvectors of A with eigenvalue λ . Conversely, an eigenvector of A in V_i has, for each ν , a coefficient vector that is either zero or an eigenvector of $F^{(i)}$. So the eigenspace of A is n_i copies of that of $F^{(i)}$. The last statement is $F_{ab}^{(i)} = \langle w_\nu^{(a)}, A w_\nu^{(b)} \rangle$. ■

Note how the work is divided. *Group theory fixes the block structure — the sizes m_i , the multiplet sizes n_i , the adapted basis — and the dynamics only fills in the small matrices $F^{(i)}$.* An

eigenvalue problem of size $\dim V$ splits into independent problems of sizes m_i . An irreducible that occurs once needs no diagonalisation at all: its $F^{(i)}$ is a number, and its adapted vectors are eigenvectors. Section 6.4 shows both cases: in its first example every $F^{(i)}$ is a number, in its second a genuine 2×2 matrix.

For a Hamiltonian, the lower bound in the next corollary needs only Proposition 6.3, even on an infinite-dimensional state space. A finite-dimensional energy eigenspace and an isotypic component are both invariant, so their intersection is too. If it is not zero, it contains an irreducible invariant subspace of type i , of dimension n_i . The second statement is the theorem itself, for a Hamiltonian on a finite-dimensional space, such as the orbitals of the next section.

Corollary 6.8 (Symmetry-enforced degeneracy). Every energy level in the isotypic component of type i is at least n_i -fold degenerate, $n_i = \dim V^{(i)}$. If the eigenvalues of each reduced matrix are simple and no two blocks share an eigenvalue, each energy level is exactly one irreducible multiplet.

Lecture 4 called a degeneracy that the symmetry dictates enforced, and promised to tell it apart from an accidental one. We can now do so precisely. A degeneracy of n_i is *enforced*. A larger one is *accidental*: an eigenvalue repeated inside one $F^{(i)}$, or two blocks that happen to share one. A generic small perturbation that keeps the symmetry removes an accidental degeneracy — unless a symmetry not yet taken into account enforces it. The dimensions of the irreducible representations are the sizes of the multiplets in a symmetric spectrum.

6.4 A particle hopping on a triangle

Put one electron orbital on each of three identical atoms at the corners of an equilateral triangle — the three hydrogens of ammonia, or the three carbons of the cyclopropenyl ring — and write $|j\rangle$ for the orbital on atom j (Figure 6.1). The electron hops between neighbouring atoms with amplitude $-t$, and on a triangle every two atoms are neighbours, so

$$H = -t \sum_{j \neq k} |j\rangle \langle k| = -t (J - \mathbb{1}) = -t \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

in the basis $|1\rangle, |2\rangle, |3\rangle$, with J the all-ones matrix of Example 6.6. The group $C_{3v} \cong S_3$ permutes the orbitals by the permutation representation, and H commutes with it, because a permutation matrix commutes with J .

By Lecture 5, $\mathbb{C}^3 \cong A_1 \oplus E$, each once. So Theorem 6.7 says that H is a number on each component, and Example 6.6 has the components: $J = 3P_{A_1}$ and $\mathbb{1} = P_{A_1} + P_E$, so

$$H = -2t P_{A_1} + t P_E.$$

The two reduced matrices are the numbers $F^{(A_1)} = -2t$ and $F^{(E)} = t$. The levels are $-2t$, on the symmetric orbital

$$u_0 = |1\rangle + |2\rangle + |3\rangle,$$

and t , doubly degenerate, on the plane V_0 . Symmetry has diagonalised H without a single matrix computation, and the double degeneracy is enforced: it is the dimension of E .

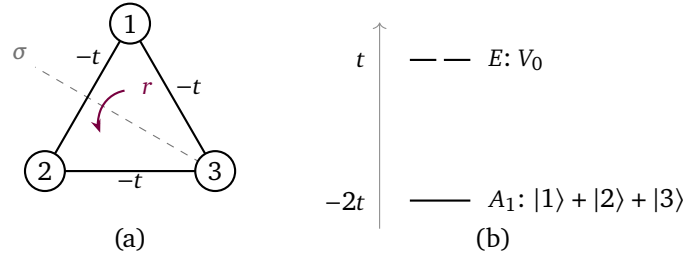


Figure 6.1: The particle on a triangle. (a) One orbital $|j\rangle$ on each atom j , and the hopping amplitude $-t$ on each bond. The rotation r takes atom 1 to the place of atom 2. The mirror σ (the dashed line) fixes atom 3 and swaps atoms 1 and 2. (b) The two energy levels, drawn for $t > 0$: the singlet A_1 at $-2t$ and the doublet E at t .

Multiplying out confirms it. Take u_0 and the vector $f_1 = |1\rangle - |2\rangle$ of V_0 :

$$Hu_0 = -t \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -t \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = -2t u_0,$$

$$Hf_1 = -t \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = -t \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = t f_1.$$

In the same way $Hf_2 = t f_2$ for $f_2 = |2\rangle - |3\rangle$, and f_1, f_2 span V_0 . These are the vectors u_0, f_1, f_2 of Lecture 4's tutorial.

Two stacked triangles. On one triangle every irreducible occurs once, so every reduced matrix is a number. A genuine matrix $F^{(i)}$ appears as soon as an irreducible occurs twice. So put a second triangle, with orbitals $|1'\rangle, |2'\rangle, |3'\rangle$, directly above the first (Figure 6.2). The electron hops with amplitude $-t$ inside each layer, as before, and with amplitude $-t'$ between an atom and the one directly above it. In the basis $|1\rangle, |2\rangle, |3\rangle, |1'\rangle, |2'\rangle, |3'\rangle$,

$$H = \begin{pmatrix} -t(J - \mathbb{1}) & -t' \mathbb{1} \\ -t' \mathbb{1} & -t(J - \mathbb{1}) \end{pmatrix}.$$

The group C_{3v} turns and reflects both layers together, and H commutes with it. Each layer carries a copy of $A_1 \oplus E$, so

$$\mathbb{C}^6 \cong 2A_1 \oplus 2E,$$

and now $m_{A_1} = m_E = 2$.

The symmetry-adapted basis is the one we already have, taken once in each layer: the copy $a = 1$ lives in the lower layer and the copy $a = 2$ in the upper one. A prime puts the same coefficients on the upper layer. For A_1 the basis is

$$w^{(1)} = u_0, \quad w^{(2)} = u'_0 = |1'\rangle + |2'\rangle + |3'\rangle,$$

and for E it is

$$w_\nu^{(1)} = f_\nu, \quad w_\nu^{(2)} = f'_\nu, \quad \nu = 1, 2.$$

Copies built from the same coefficients transform by the same matrices, as the theorem requires. The basis is not orthonormal, $\langle f_1, f_2 \rangle = -1$, so we read $F^{(i)}$ off the expansion coefficients, as in the theorem's formula, not off inner products. Inside a layer H acts as on

one triangle, and the hopping between layers carries each vector to its copy in the other layer. So

$$Hu_0 = -2t u_0 - t' u'_0, \quad Hu'_0 = -t' u_0 - 2t u'_0,$$

and, for each partner $\nu = 1, 2$,

$$Hf_\nu = t f_\nu - t' f'_\nu, \quad Hf'_\nu = -t' f_\nu + t f'_\nu.$$

The coefficients of $Hw_\nu^{(b)}$ are the b th column of the reduced matrix:

$$F^{(A_1)} = \begin{pmatrix} -2t & -t' \\ -t' & -2t \end{pmatrix}, \quad F^{(E)} = \begin{pmatrix} t & -t' \\ -t' & t \end{pmatrix}.$$

The matrix $F^{(E)}$ is the same for f_1 and for f_2 , as the theorem promises. Its eigenvectors are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, the two layers in phase, and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, the two layers in opposite phase:

$$F^{(E)} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} t & -t' \\ -t' & t \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (t - t') \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad F^{(E)} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (t + t') \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

In the same way $F^{(A_1)}$ has the eigenvalues $-2t - t'$ and $-2t + t'$.

So the 6×6 eigenvalue problem has become two 2×2 problems, and the one for E is solved once for both partners. The spectrum is two singlets $-2t \mp t'$ and two doublets $t \mp t'$, and $1 + 1 + 2 + 2 = 6$. Each doublet is enforced: an eigenvector c of $F^{(E)}$ gives the two states $c_1 f_\nu + c_2 f'_\nu$, one for each ν . Each level of one triangle splits in two (Figure 6.2). At $t' = 3t/2$ the levels $-2t + t'$ and $t - t'$ meet: two blocks share an eigenvalue, the accidental degeneracy of Section 6.3. The prism has one more symmetry, which we did not use: the mirror halfway between the layers, which swaps them. It is why both reduced matrices have the eigenvectors $\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$. With this larger group every irreducible occurs once, and no diagonalisation is left.

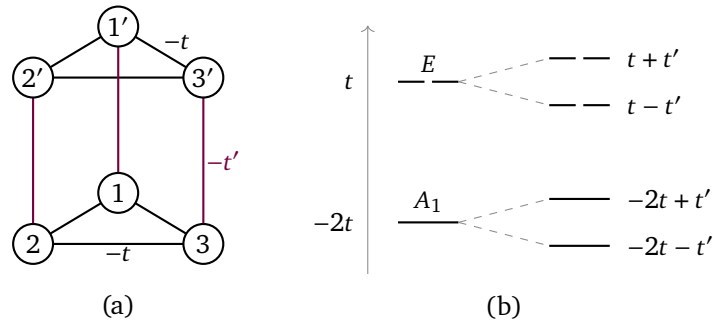


Figure 6.2: Two stacked triangles. (a) The orbitals $|j\rangle$ of the lower layer and $|j'\rangle$ of the upper layer. The hopping is $-t$ along every edge of a triangle and $-t'$ along every vertical edge. (b) On the left the levels of one triangle, on the right those of two; drawn for $t > 0$ and $t' = t/2$. Each level of one triangle splits in two, and the doublet E stays a doublet.

Back on one triangle, now keep only the rotations C_3 . Let $\omega = e^{2\pi i/3}$ and

$$v_k = \sum_{j=1}^3 \omega^{-kj} |j\rangle, \quad k = 0, 1, 2.$$

Since $\omega^{-1} = \omega^2$ and $\omega^{-3} = 1$, these are

$$v_0 = u_0, \quad v_1 = \omega^2 |1\rangle + \omega |2\rangle + |3\rangle, \quad v_2 = \omega |1\rangle + \omega^2 |2\rangle + |3\rangle.$$

The rotation sends $|j\rangle \mapsto |j+1\rangle$, so

$$\rho(r)v_1 = \omega^2 |2\rangle + \omega |3\rangle + |1\rangle = \omega(\omega^2 |1\rangle + \omega |2\rangle + |3\rangle) = \omega v_1,$$

and in the same way $\rho(r)v_k = \omega^k v_k$: the line $\mathbb{C}v_k$ carries χ_k . Each of the three characters of C_3 occurs once, so each v_k is an eigenvector of H . Since $1 + \omega + \omega^2 = 0$, the components of v_1 and v_2 sum to zero: both lie in V_0 , so their energy is t , while that of $v_0 = u_0$ is $-2t$. But χ_1 and χ_2 are different representations of C_3 , and C_3 alone does not force their energies to agree. What does?

The reflections of C_{3v} force it, and so, independently, does the reality of the Hamiltonian. First, the reflections reverse the direction round the triangle and swap the lines of v_1 and v_2 , which together make up E . The mirror σ of Figure 6.1, for one, swaps $|1\rangle$ and $|2\rangle$ and fixes $|3\rangle$:

$$\rho(\sigma)v_1 = \omega^2 |2\rangle + \omega |1\rangle + |3\rangle = v_2.$$

Since $\rho(\sigma)$ commutes with H , if v_1 has energy E , then so does v_2 : $Hv_2 = \rho(\sigma)Hv_1 = E\rho(\sigma)v_1 = Ev_2$. Second, the Hamiltonian is *real*: complex conjugation, which for a spinless particle is time reversal, commutes with H and turns v_1 into v_2 ,

$$\overline{v_1} = \overline{\omega^2 |1\rangle + \omega |2\rangle + |3\rangle} = \omega |1\rangle + \omega^2 |2\rangle + |3\rangle = v_2,$$

so the two have the same energy. The second reason is general. For a real H and real matrices $\rho(g)$, conjugation carries the isotypic component of an irreducible $V^{(i)}$ to that of its conjugate $\overline{V^{(i)}}$. Take as adapted basis of the second the conjugates of the adapted basis of the first. Then the reduced matrices obey $F^{(\bar{i})} = \overline{F^{(i)}}$, and since the eigenvalues of $F^{(i)}$ are energies, hence real, the two blocks have the same eigenvalues. So irreducibles with non-real characters come in pairs with equal energies, and the pair forms one real multiplet. A degeneracy that looks accidental for the group at hand is often enforced by reality, or by a larger group.

6.5 Degeneracy, further [self-study]

The Laplacian commutes with rotations and reflections. This proves Proposition 6.2. By the chain rule, $\partial_\alpha [\psi(g^{-1}\vec{x})] = (g^{-1})_{\beta\alpha} (\partial_\beta \psi)(g^{-1}\vec{x})$. Summing the second derivatives gives the coefficients

$$\sum_\alpha (g^{-1})_{\beta\alpha} (g^{-1})_{\gamma\alpha} = (g^{-1}(g^{-1})^\top)_{\beta\gamma} = \delta_{\beta\gamma}$$

by orthogonality. Hence $\nabla^2 \rho(g) = \rho(g) \nabla^2$. The potential term acts by multiplication, and $\rho(g)V\rho(g)^{-1}$ is multiplication by $V(g^{-1}\vec{x})$. This equals multiplication by V for all g exactly when V is invariant.

Real, complex and quaternionic irreducibles. In the language of real types (Frobenius–Schur theory, stated without proof): an irreducible is of *real type* if it is equivalent to one by real matrices, of *complex type* if it is not equivalent to its conjugate, and of *quaternionic type* if it is equivalent to its conjugate but has no real form. The three cases are told apart by $\frac{1}{|G|} \sum_g \chi(g^2) = 1, 0, -1$. An irreducible of complex type must occur together with its conjugate in a real system, as χ_1 and χ_2 of C_3 did on the triangle. The planar rotation representation of C_3 shows the same thing for vibrations: a real matrix that commutes with the rotation $R(2\pi/3)$ has the form $a\mathbb{1} + bR(\pi/2)$ with $R(\pi/2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the quarter-turn; if it is also symmetric, then $b = 0$, and so the two conjugate complex singlets form a real doublet.

Example 6.9 (The square well: forced and accidental degeneracy side by side). A particle in a two-dimensional infinite square well $(x, y) \in [-\frac{a}{2}, \frac{a}{2}]^2$ has the symmetry group D_4 of the square. The eigenfunctions factorise: $\psi_{nm} = \varphi_n(x)\varphi_m(y)$ with $E_{nm} \propto n^2 + m^2$. Here φ_n is a cosine for odd n and a sine for even n , so that $\varphi_n(-x) = (-1)^{n+1}\varphi_n(x)$.

Track the D_4 action on the pair $W_{nm} = \text{span}\{\psi_{nm}, \psi_{mn}\}$, $n \neq m$. The quarter-turn sends $\psi_{nm} \mapsto (-1)^{m+1}\psi_{mn}$, so its matrix is off-diagonal. The half-turn multiplies by $(-1)^{n+m}$. The axis reflections multiply by $(-1)^{n+1}$ and $(-1)^{m+1}$, and the diagonal reflection swaps the pair. The classes are $(e, r^2, 2r, 2\sigma_v, 2\sigma_d)$: σ_v are the two axis reflections and σ_d the two diagonal ones (the classes of s and rs in Lecture 5's table of D_4). The character of W_{nm} is therefore

$$\chi_W = (2, 2(-1)^{n+m}, 0, (-1)^{n+1} + (-1)^{m+1}, 0).$$

For $n+m$ odd this is $(2, -2, 0, 0, 0) = \chi_5$, the two-dimensional irreducible representation of D_4 . So the degeneracy of, say, E_{12} is *forced* by symmetry. For n, m of *equal* parity the character is $(2, 2, 0, \pm 2, 0)$. It decomposes into two *inequivalent one-dimensional* representations: the states $\psi_{nm} \pm \psi_{mn}$ belong to different symmetry types. Their shared energy, for instance $E_{13} = E_{31}$, is an *accidental* degeneracy of this special separable potential. A generic D_4 -symmetric perturbation of the well will split it. The well even harbours purely arithmetic accidents: $E_{17} = E_{55}$, from $1 + 49 = 25 + 25$, connects different (n, m) pairs altogether. One simple system shows all three phenomena (Figure 6.3).

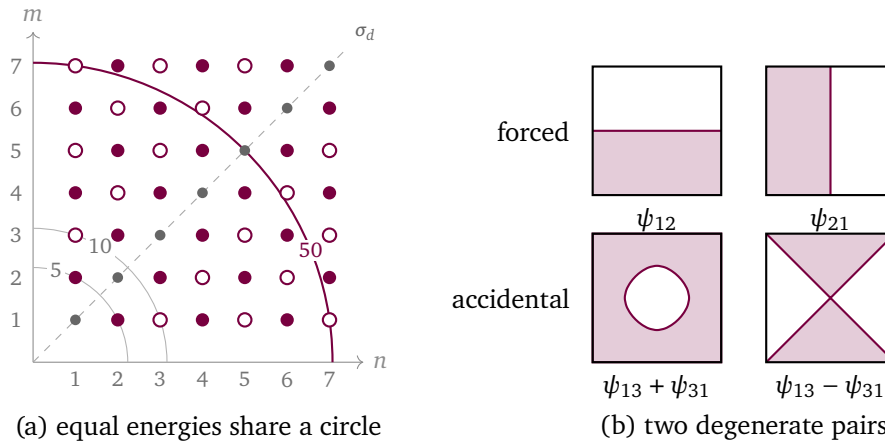


Figure 6.3: The square well of Example 6.9. (a) Each state ψ_{nm} is a lattice point, and $E_{nm} \propto n^2 + m^2$ is its squared distance from the origin, so states of equal energy lie on one circle. The diagonal reflection σ_d swaps (n, m) and (m, n) . Filled dots ($n+m$ odd) pair into a copy of the two-dimensional irreducible representation, and their degeneracy is forced. Open dots ($n \neq m$ of equal parity) pair accidentally. A grey dot on the diagonal has no partner of its own. The circle $n^2 + m^2 = 50$ meets three states, an arithmetic accident. (b) The patterns of two degenerate pairs, with nodal lines in colour. A quarter-turn carries ψ_{12} into ψ_{21} , up to sign: the two form one doublet. The combinations $\psi_{13} \pm \psi_{31}$ share an energy but look nothing alike. One keeps every symmetry of the square (χ_1 in Lecture 5's table of D_4); the other changes sign under the quarter-turn and the diagonal reflections (χ_3).

Hydrogen. The levels of the hydrogen atom are n^2 -fold degenerate (ignoring spin), far more than the $(2\ell + 1)$ -fold multiplets of the rotation group. The extra degeneracy is enforced

by a hidden four-dimensional rotation symmetry, generated by the angular momentum and the Runge–Lenz vector. A starred part of Lecture 13 touches on it.

Partner functions. The character projector P_i picks out the isotypic component but does not see inside it. When the matrices $D^{(i)}$ themselves are known, unitary, the operators

$$P_{\mu\nu}^{(i)} = \frac{n_i}{|G|} \sum_{g \in G} \overline{D_{\mu\nu}^{(i)}(g)} \rho(g)$$

resolve V_i direction by direction. On a symmetry-adapted basis of any copy of $V^{(i)}$ they act by $P_{\mu\nu}^{(i)} w_\beta = \delta_{\nu\beta} w_\mu$, and they annihilate the other types; the proof is one line from the great orthogonality theorem of Lecture 5's self-study section. So $P_{\mu\mu}^{(i)}$ projects onto the μ th direction of each type- i copy, and $P_{\mu\nu}^{(i)}$ carries the ν th partner into the μ th. Applied to a few trial vectors, they produce a whole symmetry-adapted basis, partners included.

Summary.

- *Quantum symmetry.* G is a symmetry when $[\rho(g), H] = 0$; then every energy eigenspace carries a representation of G (Definition 6.1, Proposition 6.3).
- *Projectors.* $P_i = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \rho(g)$ projects onto the isotypic component V_i , and $V = \bigoplus_i V_i$ (Theorem 6.5).
- *Block diagonalisation.* An operator commuting with G acts on V_i through an $m_i \times m_i$ matrix $F^{(i)}$, the same for every partner; its eigenvalues are at least n_i -fold degenerate (Theorem 6.7). Group theory fixes the blocks, dynamics fills them in.
- *Degeneracy.* A level of type i is at least n_i -fold degenerate, and generically exactly (Corollary 6.8); more is accidental, or a sign of reality or of a larger symmetry.
- *The triangle.* $H = -2tP_{A_1} + tP_E$: one level at $-2t$ and an enforced doublet at t . On two stacked triangles each irreducible occurs twice, and two 2×2 reduced matrices give the singlets $-2t \mp t'$ and the doublets $t \mp t'$.

Tutorial. Exercise Sheet 6.

References

Tinkham, *Group Theory and Quantum Mechanics*, Ch. 4–5 (symmetry and degeneracy); Serre, *Linear Representations of Finite Groups*, §2.6 (the canonical decomposition); Elliott–Dawber, *Symmetry in Physics*, Ch. 4.